

AIMS
HDDA
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LAB — Non-parametric regression

Kernel Estimate

1 Simulation of data

We suppose that we have the observation of n real random variables $Y_i, i = 1, \dots, n$ independent, and we place ourselves in the framework of the standard regression model

$$\begin{aligned} Y_i &= f(x_i) + \epsilon_i, \quad i = 1, \dots, n, \\ \epsilon_1, \dots, \epsilon_n &\text{ i.i.d. avec } \mathbb{E}(\epsilon_i) = 0, \quad \text{Var}(\epsilon_i) = \sigma^2, \end{aligned} \quad (1)$$

where the predictors x_i are **deterministic** and have values in $\mathbb{R}$ and f is an unknown function that we want to estimate. In this tutorial, we will use the following code to generate 4 different regression functions :

```
# Set of 4 functions
T = 1000
t = (1:T)/T

truef1 = 0.5 + (0.2*cos(4*pi*t)) + (0.1*cos(24*pi*t))
truef2 = 0.2 + 0.6*(t > 1/3 & t <= 0.75)
truef3 = 4*sin(4*pi*t) - sign(t - .3) - sign(.72 - t) + 5
truef4 = sqrt(t*(1-t))*sin((2*pi*1.05)/(t+.05)) + 0.5

# Visualisation
x11()
plot(t,truef1,type = "l", col="red", lwd=2)

# Choice of the design points : on [0,1]
n <- 100
x <- (1:n)/n

# Choice of signal noise ratio
rsnr = 5

# Data 1
f1 = 0.5 + (0.2*cos(4*pi*x)) + (0.1*cos(24*pi*x))
sigma1 <- sd(f1)/rsnr # Niveau de bruit
Y1 <- f1 + rnorm(n,mean=0,sd=sigma1)

# Visualisation
x11()
```

```
plot(x,Y1,type="p",pch=19)

# Visualisation
x11()
plot(x,Y1,type="p",pch=19)
lines(t,truef1, type = "l", col="red", lty="dotted", lwd=2)
```

Q. 1 Use the code above to view the data for each of the 4 regression functions f_1, f_4 .

In nonparametric regression, no particular model for f is assumed. Under R, a Nadaraya-Watson estimator or a local polynomial estimator can be implemented from the function `locpoly` of the library `KernSmooth`. We recall that the Nadaraya-Watson estimator corresponds to an estimator by local polynomial of degree zero.

Q. 2 Fit a kernel estimator for each of the 4 data sets $Y_1, \dots, Y_4$ which correspond to the functions $f_1, \dots, f_4$. For this you can use the function `help` of R to know the syntax of `locpoly`.

Q. 3 For each regression function, vary the window parameter h for the kernel estimator. For different values of h , also plot the point squared error the point squared error $x \mapsto (\hat{f}_h(x) - f(x))^2$. What do you find about the bias and variance of the estimators as h varies? varies?

Q. 4 What can you say about the quality of the choice of h by cross-validation for the estimation of the regression functions $f_1, \dots, f_4$?